

Reimagining Museum Engagement

Augmented Reality in Contemporary Curatorial Practice

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Abstract

This paper explores the transformative role of augmented reality (AR) in reimagining museum engagement through two case studies: the Pocketbook AR App and Echoes of Newark. Both projects leverage mobile AR to enhance visitor interaction, memory retention, and post-visit reflection. The Pocketbook app enables users to scan and collect artifacts, access contextual information, and curate personalized digital galleries. Echoes of Newark gamifies the museum experience through trivia-based scavenger hunts and animated artworks, fostering curiosity and repeat engagement. Drawing on constructivist learning theory and participatory design, the paper situates these tools within broader trends in digital heritage and curatorial innovation. It also examines their technical implementation using Unity, Vuforia, and AI animation platforms. By comparing their features and educational goals, the paper highlights synergies between the two projects and discusses how they support emerging curatorial practices that blend physical and digital storytelling, expand access, and redefine the museum experience in the 21st century.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing design and evaluation methods; Empirical studies in interaction design; • **Applied computing** → Interactive learning environments.

Keywords

Augmented reality, Photogrammetry, Interactive applications, Digital heritage, Generative AI

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1 Introduction

The integration of immersive technologies into museum environments is reshaping how audiences engage with cultural heritage by reinventing museum service models, focusing on visitor satisfaction and accessibility [Trunfio et al. 2022]. As institutions seek to remain relevant in a digital-first world, tools like augmented reality (AR), virtual reality (VR), and interactive mobile applications are emerging as powerful mediums for storytelling, education, and individualized engagement [Boboc et al. 2022] [Hincapié et al. 2021]. This paper examines two innovative projects—the Pocketbook AR App and Echoes of Newark—that exemplify this shift. Both projects aim to enhance visitor interactions, extend the museum experience beyond physical walls, and support new curatorial strategies.

The paper explores how these tools address challenges such as declining attention spans, limited accessibility, and the need for personalized learning. It also investigates how curators and educators can leverage AR to create hybrid experiences that blend physical and digital artifacts. Through presented case studies, it investigates the following research questions: (1) How do AR technologies reshape visitor engagement? (2) What curatorial opportunities emerge from mobile and immersive experiences? And (3) How do these tools support memory, learning, and access? By analyzing the design, implementation, and impact of these two projects, the paper contributes to broader discussions on the future of digital heritage and participatory museum practice.

2 Contextual Framework

The use of digital technologies in cultural heritage has been widely studied, with scholars emphasizing their potential to democratize access, enhance learning, and preserve intangible heritage. Augmented reality, in particular, has been recognized for its ability to overlay interpretive content onto physical artifacts, creating layered experiences that engage multiple senses. Additionally, AR facilitates individualized experiences in group settings, allowing for hybrid content interactions and authorship. Research by Geroimenko [Geroimenko 2021] highlights how AR fosters deeper emotional connections and supports constructivist learning models in museum settings by extending their reach, personalizing learning, and creating hybrid physical-digital experiences. Zhang et al. [Zhang et al. 2024] review explores how AR and VR are used in hybrid museum exhibitions to enhance storytelling and visitor interaction. It provides a conceptual framework for understanding how immersive technologies support curatorial innovation and audience engagement—core themes in both case studies. Storytelling and user authorship emerge as a critical dimension of emerging practices, as visitor engagement and attention span remain a central

concern in museum studies. The Metropolitan Museum of Art study showed that the average visitor spends less than 30 seconds viewing a single artwork (mean viewing time 27.2 seconds, the median 17.0 seconds, and the mode of 10.0 seconds), often leaving with limited understanding or memory [Smith and Smith 2001]. This study is widely cited and forms the empirical basis for many discussions on museum engagement and attention span. Similar findings were demonstrated by [Lanir et al. 2017], who used automatic tracking to analyze visitor movement and interaction patterns in museums, noticing that most visitors spend only brief moments at each exhibit, often without engaging deeply. Digital interventions—such as mobile guides, interactive kiosks, and AR overlays—have been proposed as solutions to this problem. At the same time, curators are increasingly adopting digital tools not only for exhibit planning but also for interpretation, audience analytics, and remote access. This paper situates the two case studies within this evolving landscape, drawing connections between technological innovation and curatorial transformation.

2.1 Learning Frameworks in Digital Heritage

The convergence of educational theory and digital innovation is reshaping the landscape of museum curation. At the heart of this transformation lies the integration of constructivist learning theory, Bloom's taxonomy, and Kolb's experiential learning cycle—each offering a powerful lens through which digital heritage initiatives can be designed to foster deeper engagement and understanding. Constructivist Learning Theory posits that learners actively construct knowledge through experience and reflection. In the context of digital heritage, this theory supports the design of interactive exhibits and virtual environments where visitors are not passive recipients but co-creators of meaning. For instance, augmented reality (AR) reconstructions of historical sites or participatory digital archives allow users to explore, question, and interpret cultural narratives, aligning with the constructivist emphasis on learner agency and contextual understanding [Hein 2002]. Along similar lines, Bloom's Taxonomy provides a hierarchical framework for cognitive development, from basic knowledge recall to higher-order thinking skills like analysis, synthesis, and evaluation. Digital heritage platforms can be structured to guide users through these levels. A virtual museum tour might begin with factual information (Remembering), followed by interactive timelines or quizzes (Understanding and Applying), and culminate in user-generated content or debates on cultural significance (Analyzing, Evaluating, Creating). This scaffolding ensures that digital curation not only informs but also challenges and inspires critical engagement [Dixit and Thomas 2021].

Together, these frameworks advocate for a museum curation model that is participatory, reflective, and intellectually rigorous. By embedding constructivist principles, cognitive scaffolding, and experiential cycles into digital heritage initiatives, curators can create inclusive spaces where diverse audiences not only access culture but actively shape its meaning.

3 Tools and Techniques

This shift in curatorial practices and visitor expectations is driven by emerging digital technologies allowing for high-fidelity 3D scanning, real-time interactions, and generative AI-based individualized content. Photogrammetry is a foundational technique in digital heritage that enables the creation of highly detailed 3D models from photographs, integrating real-world cultural artifacts into AR and VR environments with the use of game engines. In the context of art and architecture, photogrammetry allows museums and preservationists to digitize sculptures, paintings, and historic buildings with remarkable accuracy [Nagakura and Sung 2017] [Kepczynska-Walczak et al. 2024] [Kepczynska-Walczak-Anetta 2025]. These digital replicas can then be explored in AR—where users overlay them into their physical space—or in VR, where entire historical environments can be reconstructed for immersive storytelling and education [Güleç et al. 2016] [Di Mascio 2017] [Nagakura and Sung 2014] [Di Mascio 2020].

This technology is particularly impactful in preserving and sharing cultural heritage that is fragile, remote, or no longer physically accessible. In museums, photogrammetry supports both public engagement and curatorial planning, enabling curators to visualize exhibit layouts and audiences to explore artifacts in new, interactive ways. To enhance this interactivity, it is often combined with game engines, such as Unity or Unreal, to achieve personalized gameplay and a more engaging (gamified) experience. The personalization can be further reinforced with multimodal AI, particularly large language models (LLMs), which automate the generation of interpretive content for 3D-scanned artifacts created via photogrammetry. For example, once a sculpture is digitized, AI can generate multilingual descriptions, historical context, or even interactive narratives tailored to the visitor's age, interests, or learning goals. When integrated into AR, this content can be overlaid in real time as users explore exhibits, creating a personalized and immersive educational journey. AI can also power adaptive interfaces that respond to user behavior—such as gaze tracking or dwell time—offering deeper insights or guiding users to related artifacts. AI-driven analytics can also process visitor interaction data—such as which artifacts are most frequently scanned or revisited—to inform exhibit design and layout.

4 Case Study 1: Pocketbook AR App

In response to the challenges facing modern cultural institutions—from declining attendance among younger generations to the ephemeral nature of the traditional visit—the "Interconnected Experience" project by Kamil Ciemierkiewicz and Hee Jun proposes a sophisticated and holistic solution. This augmented reality (AR) application, designed for the Newark Museum of Art, is more than just a digital guide; it is a comprehensive ecosystem that redefines the relationship between the visitor, the collection, and the institution itself. By transforming the museum visit into a persistent and portable experience, the project tackles the core problem of superficial engagement and empowers every visitor to become the curator of their own personal virtual collection (figure 1).

The project aspires to combat the "overwhelmed" feeling many visitors face and, at the same time, to mitigate the "30-second gaze" phenomenon by addressing the problem of the fleeting museum

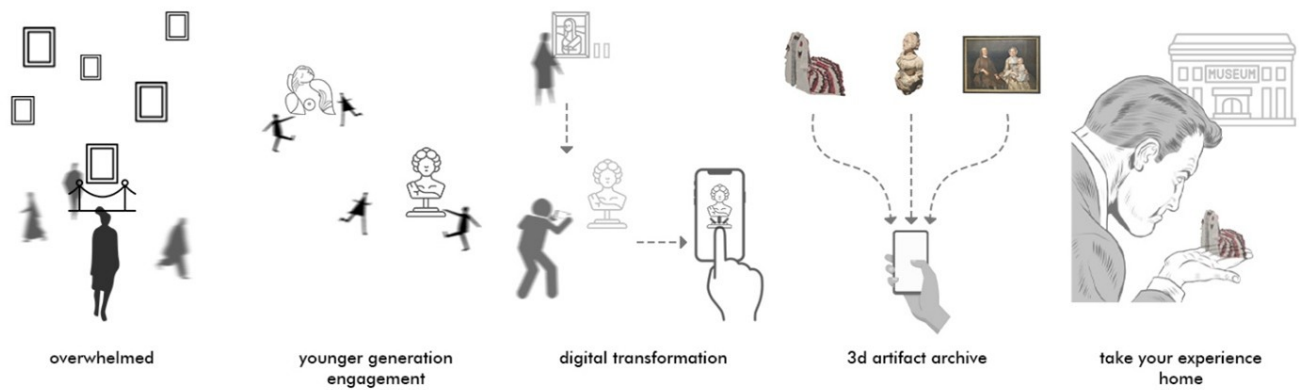


Figure 1: Conceptual framing and drives of the Pocketbook AR App.



Figure 2: Photogrammetric assets placed within an AR app with a pass-through view of the gallery mimics the “curatorial mode”.

memory. It asks the critical question: “What do you leave with?” after a cultural site visit. Instead of a cluttered camera roll of photos, the project aspires to provide visitors with a lasting, interactive “pocketbook”—a digital archive of the art that captivated them. This portable experience is intended not just to be taken home, but to be taken “anywhere,” making the museum’s collection a persistent part of the visitor’s life and fostering a lifelong interest in art and culture. It is perhaps a contemporary (Netflix) version of a library or a video store where one could check out, instead of a book or a movie, a virtual art piece from the museum collection (figure 2).

4.1 Gameplay

The “Interconnected Experience” is a multi-faceted AR application with distinct modes for visitors and museum professionals. For the visitor, the journey begins inside the museum. Upon encountering an artwork, the user can scan it with their device. This action unlocks a rich menu of interactive options: they can explore a high-fidelity 3D model of the object in “3D Orbit,” listen to a curated audio narrative, or watch supplementary videos (figure 3). As visitors explore the exhibition, they can “like” or collect (check it out) artifacts, adding them to their personal digital archive. This collection is the core of the post-visit experience. Once they leave the museum, users can access their “pocketbook” at any time. Through the app’s “Remote Access” feature, they can use AR to place these

virtual artifacts in their own environment—placing a sculpture on their coffee table or hanging a painting on their living room wall (figure 4) The second key function is the "Curator Mode," an institutional tool that allows museum staff to use the same AR technology to virtually place 3D models of artworks within a physical gallery space, enabling them to design and experiment with exhibition layouts efficiently and without risk to the artifacts.

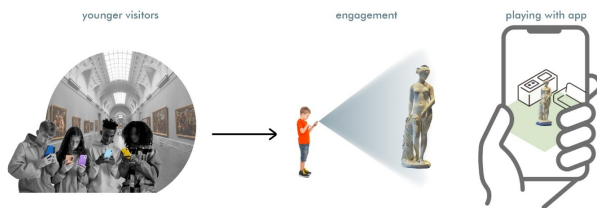


Figure 3: User gameplay showing individual steps.



Figure 4: Remote access allows for enjoying museum exhibits from any place and at any time.

4.2 User Experience

The user journey is designed to be intuitive, inclusive, and engaging. Upon entering the museum, visitors are guided through a series of interactive touchpoints where they scan collection artefacts, view photogrammetric models, listen to audio guides, and watch short videos. The app supports learning at one's own pace, making it suitable for children, students, and adults alike. The core loop—scan, learn, interact, collect—is simple and satisfying. It creates a "non-traditional" path to learning that is self-paced, deeply engaging, and personally resonant. A key feature is the ability to "orbit" around 3D assets, offering perspectives that are not possible in traditional displays. The customizable room at the end of the experience allows users to curate their own gallery, reinforcing a sense of ownership and creativity. The app also avoids the need for bulky equipment, making it more accessible and user-friendly. Inside the museum, the UX is focused on discovery and learning. It provides immediate access to layered information, allowing visitors to dive as deep as they wish into an artwork's history and context. Outside the museum, the UX becomes centered on creativity and personal reflection. Users are no longer just learning; they are curating. The act of placing a virtual Rodin in one's garden or creating a customized virtual room with collected favorites fosters a powerful sense of ownership and a lasting personal connection to the art (figure 5,6).



Figure 5: Game tree (in the museum).

4.3 Technology and Methodology

The technological backbone of the "Interconnected Experience" is a robust combination of 3D scanning, 3D object recognition, and augmented reality. The creation of the app's primary assets—the "Scans of Artifacts"—relies on photogrammetry, creating geometrically accurate "digital twins" of sculptures and paintings that are essential for both close-up study and realistic AR placement. These 3D assets, along with supplementary audio and video files, are integrated into an AR application built within the Unity engine with the Vuforia plugin facilitating 2D and 3D object recognition. The Android phones were used for the deployment while GitHub repository and LLM (ChatGPT) were used for assistance with coding and problem-solving, reflecting a modern, agile, and pragmatic approach to development.

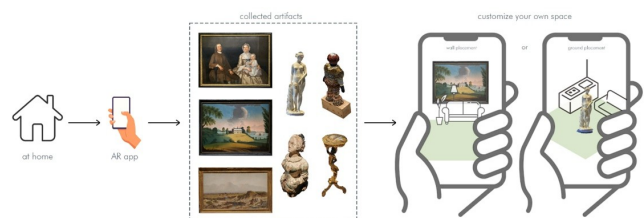


Figure 6: Game tree (remote access).

4.4 Impact and Intellectual Contribution

The most significant intellectual contribution of the "Interconnected Experience" is its holistic and dual-purpose design, which benefits both the public and the institution. The project's impact is not just in enhancing the visitor experience, but in providing a practical, integrated tool for the museum itself. For the public, the interactive "pocketbook" provides a solution for the ephemeral museum visit. It makes the experience persistent, portable, and personal, effectively democratizing the act of curation. The ability to create a lasting, interactive archive of one's favorite art fosters a level of engagement that a single visit can rarely achieve. For the institution, the "Curator Mode" offers a tangible return on investment, streamlining exhibition design and reducing the labor and risk associated with physical

installation planning. This symbiotic approach—where the same core technology serves both public engagement and professional practice—is the project's most powerful and forward-thinking contribution, offering a truly interconnected model for the museum of the future.

5 Case Study 2: Echoes of Newark

The Echoes of Newark project aspires to transform the Newark Museum of Art into a more accessible, engaging, and educational space by integrating augmented reality (AR) into the visitor experience. The project bridges the gap between traditional museum practices and contemporary digital expectations, particularly among younger and more tech-savvy audiences. By turning art exploration into an interactive scavenger hunt, the project aims to foster curiosity, encourage repeat visits, and deepen cultural engagement. It also seeks to amplify the museum's community impact by connecting its internal collections with public monuments and city landmarks, extending the museum's reach beyond its physical walls (figure 7).

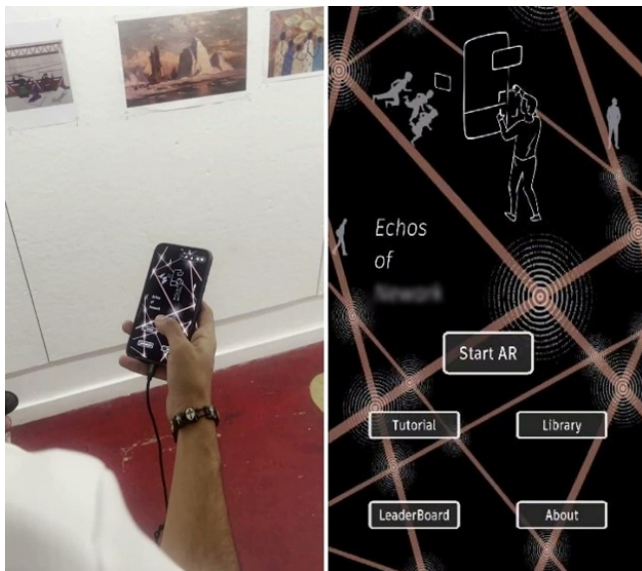


Figure 7: In-game experience of the prototype testing (left) and the UI for the opening screen (right).

5.1 Gameplay

Upon entering the museum, users scan an art object to receive their first clue. They then navigate the galleries to locate specific artworks or artifacts, guided by hints and trivia questions. Correct answers unlock new clues and add the artifact to the user's digital collection, which includes contextual information and AI-generated animated visualizations (figure 8). The experience culminates after a set minimum number of art interactions, revealing a daily leaderboard. This gamified structure not only encourages exploration but also creates a sense of progression and achievement.

5.2 User Experience

The user journey is designed to be intuitive and rewarding. Visitors interact with the museum through their smartphones or tablets, scanning artworks to trigger AR overlays and questions related to the given artefact. There are three knowledge levels for each artifact: beginner, intermediate, and advanced. This allows for fine-tuning the museum experience to the level and interest of the visitor. The nature of these questions spans from simply recalling artefact information, or itself, to analytical and interpretive questions that ask players to abstract or synthetically approach provided information. At the advanced level, some of the questions delve into general cultural knowledge that is not part of the exhibition presentation. The differentiation of question types and difficulty levels facilitates the adaptive nature of the curatorial experience based on user familiarity or interests. It allows for multiple and more frequent museum visits, even with the same exhibition. Furthermore, each correct question response enhances the player's digital collection and provides them with a virtual award. This includes AI-animated artwork providing an alternative take on its aesthetics, composition, different styles, or seasonal variations, offering a fresh perspective on familiar pieces. Both the question and reward systems are designed to pause visitors in their museum visit to take time to analyze and interpret art pieces, to overcome a 30-second attention span usually afforded in this context. Treasure hunt point of interest (POI) randomization diversified possible museum routes, further contributing to individualized user experience. Finally, a daily leaderboard fosters friendly competition that facilitates both individual and collective activities.

5.3 Technology and Methodology

The project leverages a robust technical stack (figure 9,10). Unity and the Vuforia engine are used for AR tracking and image recognition, enabling real-time interaction with physical artworks. Unity provides a potent game engine with easy-to-implement interactivity and low-to-intermediate level coding requirements. MySQL database records scores and could be extended to group competitions as well as for data collection for museum curatorial staff. AI tools, such as Clippy and PixVerse, generate animated versions of paintings, transforming static images into dynamic, stylistically varied interpretations. ChatGPT was employed to assist with Unity coding tasks (CSharp), including scene randomization and leaderboard logic. The user interface was developed using Adobe tools and Unity's UI system, with a focus on clarity, accessibility, and responsiveness. The app's architecture supports both in-museum and remote access, allowing users to build and interact with their collections from any location. The off-location functionality allows users to interact with the web-based and printed collection catalogs. It could be done in lieu of the in-person visit, or as a pre- and post-visit experience.

5.4 Impact and Contribution

Echoes of Newark contributes to the evolving discourse on digital heritage and museum innovation. It exemplifies how AR and AI can be used not just to digitize collections, but to reimagine how audiences engage with them. The project's integration of gamification, personalization, and community mapping reflects a shift toward



Figure 8: Frames from AI-animated paintings with new details and changing compositions.

participatory and experiential curation. It also demonstrates how museums can use digital tools to remain relevant and inclusive in a rapidly changing cultural landscape. By combining technical innovation with educational intent, *Echoes of **** offers a compelling model for future museum experiences—one that is interactive, memorable, and deeply rooted in place.

6 Synergies Between the Two Projects

Both the digital *Pocketbook* and *Echoes of Newark* share a common goal: to deepen visitor engagement through personalized, interactive, and portable experiences. While the *Pocketbook* emphasizes post-visit reflection and curatorial utility, *Echoes of Newark* focuses on in-the-moment exploration and gamification. Together, they represent complementary approaches to rethinking the museum experience in the digital age.

Conceptually, both projects are grounded in constructivist learning theory, which posits that knowledge is actively constructed through experience. They also align with participatory museum models that prioritize user agency, personalization, and co-creation. Technologically, leverage AR mobile platforms and AI animation to create immersive environments. Their shared use of Unity and Vuforia suggests a scalable framework that could be adapted across institutions as well as broad curatorial content types. The synergy between these projects lies in their ability to bridge physical and digital spaces, offering a holistic model for 21st-century museum engagement.

7 Emerging Curatorial Practices

The rise of AR and mobile technologies is prompting a shift in curatorial practice from static presentation to dynamic, data-informed

storytelling. Curators are no longer limited to physical space; they can now design hybrid exhibitions that blend tangible artifacts with digital overlays. Strategies like those developed in the discussed project allow curators to test exhibit layouts virtually, reducing setup time and improving spatial coherence. They also enable curators to track user interactions, providing insights into visitor behavior and preferences. Additionally, gamified and mobile platforms naturally lend themselves to a potent data collection tool that can help shape future exhibitions by correlating visitor interests with building spaces and individual art locations.

Moreover, these technologies support more inclusive and responsive curation. By preserving access to past exhibits and enabling remote participation, curators can reach broader audiences and adapt content to diverse needs. Ethical considerations—such as cultural sensitivity, digital authenticity, and accessibility—remain critical, but the potential for innovation is vast. As museums embrace these tools, curators are becoming experience designers, educators, and technologists, redefining their roles in the digital heritage ecosystem.

8 Conclusion

The *Pocketbook* AR App and *Echoes of Newark* exemplify how immersive technologies can transform museum engagement. By combining interactivity, personalization, and portability, these tools address longstanding challenges in visitor engagement and curatorial practice. They also open new pathways for learning, memory-making, and cultural access. As museums continue to evolve in response to technological and societal shifts, projects like these offer a blueprint for the future—one where art is not only seen but experienced, shared, and remembered.

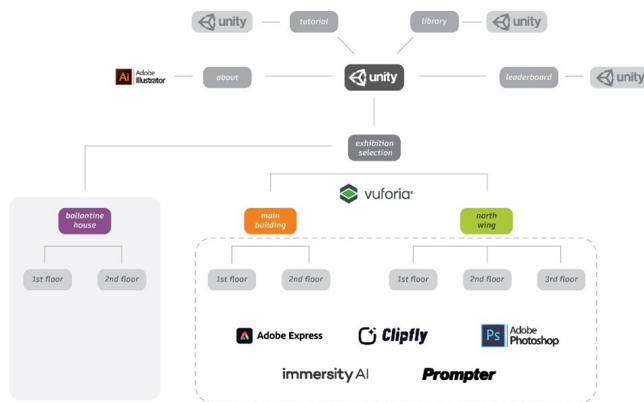


Figure 9: Game tree with technology/software used throughout the app.

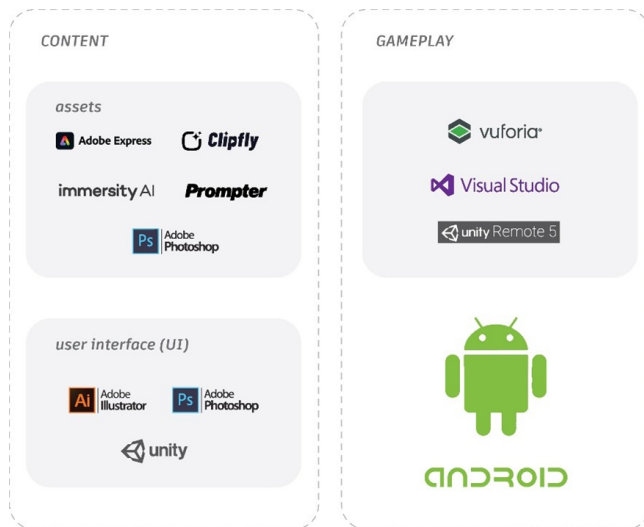


Figure 10: Software used throughout the app per development type.

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